Nagomi Numbers : Mathematical Harmony Across Multiple Dimensions

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Abstract

We introduce the Nagomi Numbers, a pair of integers $\{753, 5271\}$ exhibiting remarkable harmonic properties rooted in their shared prime factorization structure. The name "Nagomi" (harmony and tranquility) is phonetically encoded in 753: na (7), go (5), mi (3).

The fundamental relationship between these numbers is expressed by the exact division: $5271 \div 753 = 7$. Beyond this core property, we discover that both 753 and 5271 share a common factorization core involving the prime 251: specifically, $753 = 3 \times 251$ and $5271 = 3 \times 7 \times 251$. This shared structure manifests in multiple harmonic properties: the digits of both numbers sum to 15, their product $3,969,063$ contains only the digits $\{0, 3, 6, 9\}$, and all four arithmetic operations (sum, difference, product, quotient) yield results expressible as combinations of the primes $3, 7$, and 251. Furthermore, the key numbers in this system—the quotient 7, the prime factor 251, and 5271 itself—are all composed of the same digit set $\{1, 2, 5, 7\}$. These interconnected harmonic properties demonstrate how numerical beauty can manifest across multiple dimensions simultaneously.

1 Introduction

The discovery of what we call "Nagomi Numbers" began with a simple observation: the number of weeks between two personally significant dates happened to be exactly 753. This coincidence sparked our curiosity—was there something special about this number? As we explored 753 further, we discovered a remarkable related number: 5271. The relationship between

these numbers reveals a beautiful structure that extends far beyond their initial connection.

The name "Nagomi" comes from the Japanese concept of harmony and tranquility. Interestingly, the number 753 itself encodes this word phonetically in Japanese: *na* (7), *go* (5), and *mi* (3). While this linguistic connection is a charming coincidence, the true mathematical harmony of these numbers lies in their structural properties.

The fundamental relationship is elegantly simple: $5271 \div 753 = 7$. However, this division is merely the surface of a deeper harmony. When we examine the prime factorization of these numbers, we find that they share a common core: $753 = 3 \times 251$ and $5271 = 3 \times 7 \times 251$, where 251 is prime. This shared factorization structure gives rise to multiple harmonic properties that manifest across different mathematical operations and digit patterns.

In this paper, we explore the remarkable properties of the Nagomi Numbers and demonstrate how their harmonic structure extends through all four arithmetic operations, digit compositions, and factorization patterns. We show that this harmony is not merely coincidental but emerges from the fundamental role of the primes $3, 7$, and 251 in defining these numbers.

2 Definition

We define the Nagomi Numbers as the specific pair of positive integers:

$$\{753, 5271\}$$

These numbers exhibit a fundamental harmonic relationship expressed through their quotient:

$$5271 \div 753 = 7$$

This relationship can be equivalently stated as:

$$5271 = 753 \times 7$$

The harmonic properties of the Nagomi Numbers emerge from their shared prime factorization structure:

$$753 = 3 \times 251$$
$$5271 = 3 \times 7 \times 251$$

where 251 is prime.

All properties and theorems presented in this paper refer specifically to this pair of numbers and the mathematical structures they generate through arithmetic operations and factorization patterns.

3 Main Results

Theorem 3.1 (Fundamental Relationship)**.** *The Nagomi Numbers satisfy the exact division:*

$$5271 \div 753 = 7$$

Proof. Direct computation: $753 \times 7 = 5271$. □

Theorem 3.2 (Prime Factorization Structure)**.** *The Nagomi Numbers share a common prime factorization core:*

$$753 = 3 \times 251$$
$$5271 = 3 \times 7 \times 251$$

where 251 *is prime.*

Proof. By direct factorization. The primality of 251 can be verified by testing divisibility by all primes up to $\sqrt{251} \approx 15.8$. □

Theorem 3.3 (Digit Sum Harmony)**.** *Both Nagomi Numbers have the same digit sum:*

$$7 + 5 + 3 = 15$$
$$5 + 2 + 7 + 1 = 15$$

Proof. Direct computation. □

Theorem 3.4 (Product Digit Pattern)**.** *The product of the Nagomi Numbers yields a number whose digits are exclusively from the set* $\{0, 3, 6, 9\}$*:*

$$753 \times 5271 = 3,969,063$$

Proof. Direct computation. The resulting digits are $\{3, 9, 6, 9, 0, 6, 3\}$. □

Theorem 3.5 (Arithmetic Operations Factorization)**.** *All four arithmetic operations on the Nagomi Numbers yield results expressible as combinations of the primes* $3, 7,$ *and* 251*:*

- ***Sum:*** $753 + 5271 = 6024 = 2^3 \times 3 \times 251$
- ***Difference:*** $5271 - 753 = 4518 = 2 \times 3^2 \times 251$
- ***Product:*** $753 \times 5271 = 3,969,063 = 3^2 \times 7 \times 251^2$
- ***Quotient:*** $5271 \div 753 = 7$

Proof. By direct computation and factorization. □

Theorem 3.6 (Digit Set Harmony)**.** *The quotient* 7*, the prime factor* 251*, and the number* 5271 *all share the same digit set* $\{1, 2, 5, 7\}$*:*

- $7 \to \{7\}$
- $251 \to \{2, 5, 1\}$
- $5271 \to \{5, 2, 7, 1\}$

Proof. By direct observation of digit composition. □

4 Discussion

The Nagomi Numbers demonstrate a remarkable multi-layered harmony that manifests across several mathematical dimensions simultaneously. At the foundation lies the shared prime factorization structure involving $3, 7$, and 251, which generates cascading harmonic properties throughout all arithmetic operations.

4.1 Structural Harmony

The factorization structure $753 = 3 \times 251$ and $5271 = 3 \times 7 \times 251$ reveals that these numbers are built from the same fundamental primes. This shared foundation is not merely a static property but actively propagates through arithmetic operations.

4.2 Digit-Level Harmony

Beyond the factorization structure, the Nagomi Numbers exhibit harmony at the digit level. Both numbers sum to 15, and their product contains only digits from $\{0, 3, 6, 9\}$—all multiples of 3.

4.3 Multiple Manifestations of Unity

What makes the Nagomi Numbers particularly striking is how multiple forms of harmony coexist: linguistic, arithmetic, factorization, and digit patterns. Each layer reinforces the others.

4.4 Cultural Context

The number 753 holds cultural significance in Japan as "Shichi-Go-San" (Seven-Five-Three). This adds an additional layer of meaning to these numbers' designation as "Nagomi."

4.5 Future Directions

The properties of the Nagomi Numbers raise several questions: Are there other number pairs that exhibit similar multi-layered harmonic structures? What is the frequency of such harmonic pairs?

4.6 Summary of Discussion

In summary, the Nagomi Numbers serve as a compelling example of how mathematical beauty can emerge from the interplay of prime factorization, digit patterns, and arithmetic operations. Their harmonic properties demonstrate that numerical elegance is not confined to a single mathematical aspect.

5 Conclusion

We have introduced the Nagomi Numbers $\{753, 5271\}$ and demonstrated their remarkable harmonic properties across multiple mathematical dimensions. These numbers exhibit a coherent structure rooted in their shared prime factorization involving $3, 7$, and 251, which propagates through all four arithmetic operations.

The key findings include:

1. the fundamental relationship $5271 = 753 \times 7$,
2. the shared digit sum of 15,
3. the product yielding only digits from $\{0, 3, 6, 9\}$,
4. all arithmetic operations factoring through the primes $3, 7$, and 251, and
5. the self-referential digit set $\{1, 2, 5, 7\}$ appearing in the quotient, prime factor, and the Nagomi Number 5271.

These properties collectively demonstrate that the Nagomi Numbers are not merely an arbitrary pair of integers but a mathematically coherent object

exhibiting harmony across factorization, digit patterns, and arithmetic operations. The multi-layered nature of this harmony—simultaneously linguistic, structural, and computational—makes these numbers a compelling example of numerical beauty.

The discovery began with a simple observation about the number of weeks between two dates, reminding us that mathematical elegance can emerge from everyday contexts. The Nagomi Numbers stand as a testament to the idea that profound mathematical structures may be hiding in plain sight, waiting to be discovered through curiosity and careful exploration.

References

[1] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.

[2] K. H. Rosen, *Elementary Number Theory and Its Applications*, 6th ed., Addison-Wesley, 2011.

[3] D. M. Burton, *Elementary Number Theory*, 7th ed., McGraw-Hill, 2010.